Shiretechnik Solutions Pvt Ltd

SHape Your Idea To Reality

Case Study on
Busbar Thermal Design

INTRODUCTION

Busbar based Power Distribution System

offers a number of significant advantages for designers and industrial control panel users

- Simplified power distribution
- Enhanced protection
- Ease of Installation & Maintenance
- Cost-efficiency
- Space optimization
- Lower resistance /impedance than cables
- Compliant with international standards & norms

Key factors considered for busbar design are

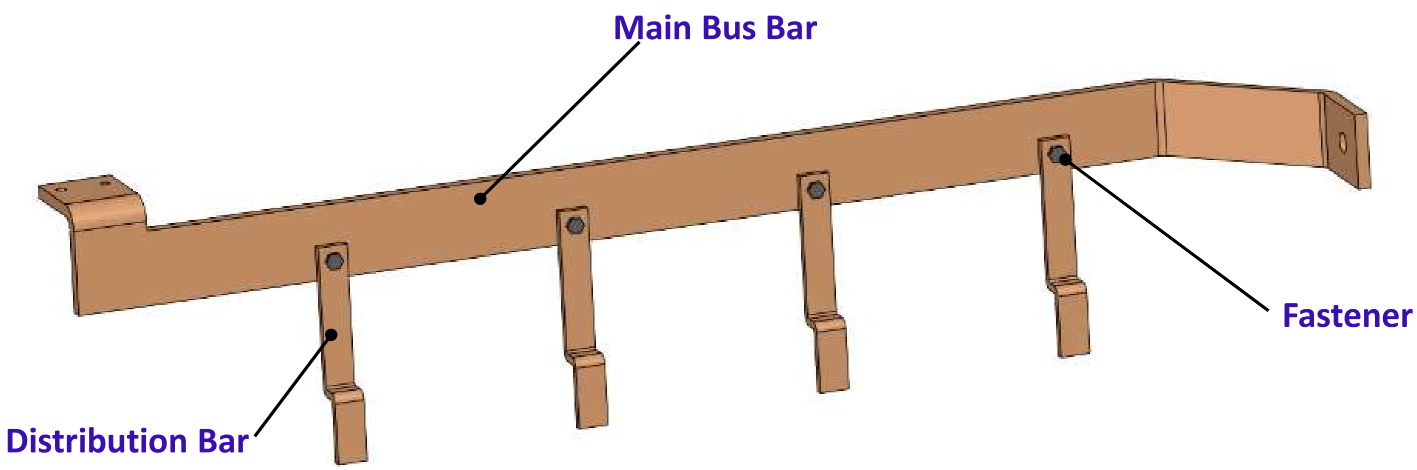
- Rated Current / Voltage / Short time current / Frequency / Insulation Level
- Operating temperature
- Ambient temperature
- Voltage level
- Short circuit current
- Materials
- Sizing

D-Node Power Distribution System

SHIRETECHNIK

Objective of Thermal Analysis

- Achieving the thermal performance per the defined design goals
- Sizing the distribution bars in accordance with thermal & electrical requirements
- Evaluation of design space for the cooling solution

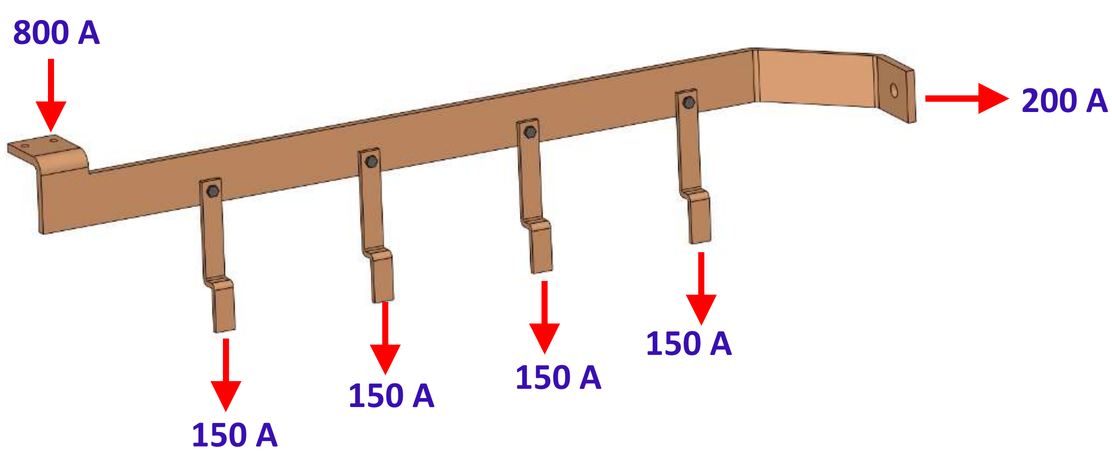


D-Node Bus Bar Power Distribution Unit

D-Node Power Distribution System

Simulation Key Inputs & Conditions

- Supply Current: 800 A
- Current per Distributed Bar: 150 A
- Ambient Temperature: 40˚C
- Busbar Max Operating Temperature: 90˚C
- Cooling Mode: Natural Cooling (Convection & Radiation)
- Busbar Material: ETP (Electrolytic Tough Pitch) Copper
- Busbar Plating: Nickel Plated
- Fasteners: Steel
- Analysis Type: Steady State

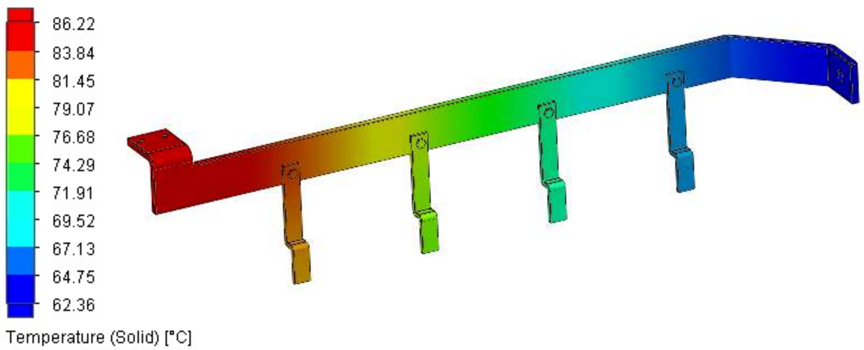


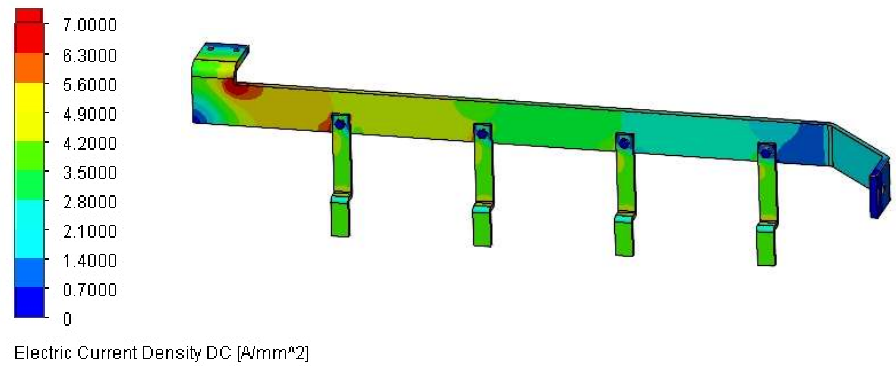
CASE STUDY

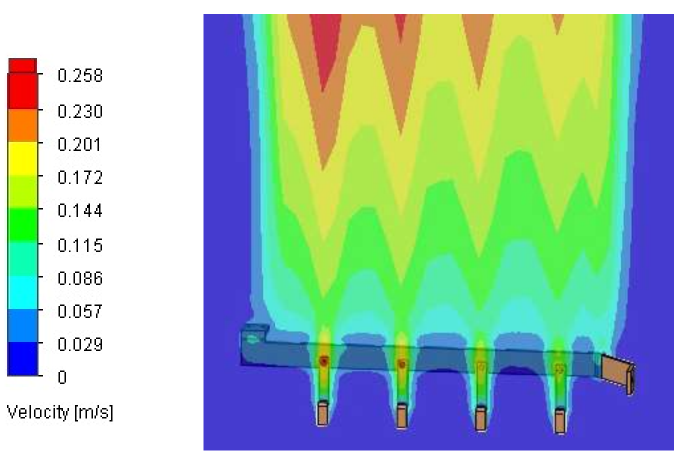
D-Node Power Distribution System

SHIRETECHNIK

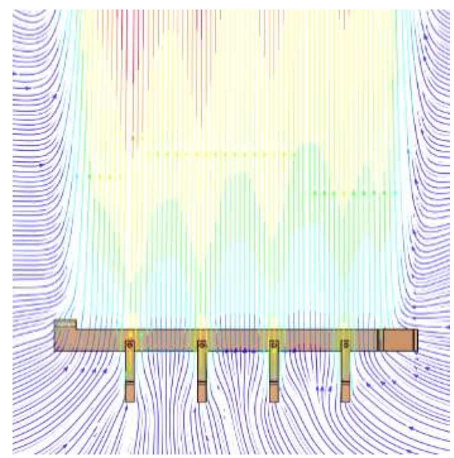
Simulation Results







Contour Plot



Streamline Plot

SHIRETECHNIK

Achievements From Simulation Analysis

- Maximum Temperature of Busbar reached to 86.2˚C (< 90˚C limit) & dissipating the total heat of 25.9 W (14.2W thru’ convection & 11.7W thru’ radiation).
- Overall cooler thermal profile
- Solution as per the thermal and electrical requirements / design goals

-END OF DOCUMENT-

Contact Shiretechnik Solutions

Email ID: contact@shiretechnik.com
Telephone: +91 80 23415100
Office Address: No. 166, 5th Main, K.E.B. Layout, Sanjaynagar, Bangalore 560 094
(Opposite to Tata Communications)